

A Comparative Study of an Individual and Ensemble Machine Learning Models for Brain Stroke Risk Prediction

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Abstract— Stroke is a problem for peoples health all around the world and it is one of the main reasons for death and long-term disability. If we can predict who is at risk of having a stroke then we can help to prevent it and get people the medical help they need on time. With more and more healthcare data available machine learning is showing a lot of promise in finding people who are at high risk of having a stroke based on their health and lifestyle.

This study focuses on comparison between individual and ensemble machine learning models for predicting brain stroke. I used a dataset that's available to the public and has information about peoples demographics and symptoms. I looked at the data to understand what it means and how different things are related to each other. I then prepared the data for use by making sure all the features are on the scale and encoding categorical features. I also made sure to leave out a feature that could give away the answer and make the models not work well.

I tried out a few machine learning models. Logistic Regression, Decision Tree, Random Forest and Gradient Boosting. To see how well they work. We looked at how each model did by checking its accuracy, precision, recall, F1-score and ROC-AUC and we also used five-fold cross-validation to make sure the models work well in different situations. We also tried combining the strengths of selected individual models using a soft Voting Classifier. What we found out was that while Logistic Regression was really good at recalling who was at risk and had a high F1-score while the Voting Classifier that is Ensemble Model was consistently performing better at predicting who was at risk and worked well in different situations. So I chose the ensemble model as our final model. We then used this model to predict stroke risk, for patients we had not seen before. The selected model is able to give two types of results that is simple yes or no, also it gives probability of stroke risk. This study shows how machine learning models can be useful in predicting stroke risk in early stages especially with ensemble model. This can help doctors to take care of patients who are diagnosed with stroke or who are at the early stage of having it and give treatment on time.

Index Terms— Stroke Prediction, Machine Learning, Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Ensemble Learning, Voting Classifier.

I. INTRODUCTION

Brain stroke is serious health problem. This happens when the brain does not get required amount of blood. Many people die or become disabled because of this. Sometimes this stroke can cause long-term issues to a person such as difficulty in making any movements, talking, thinking. This can really affect a person's daily life and their happiness.

There are many thing which can increase risk of stroke. These things include age that is how old or young someone is, if there are any hearth or breathing related problem, individual person's lifestyle. So it is very important to get to know who is at a risk of having a stroke as early as possible. This will help preventing stroke and it's very bad effects.

Traditionally, doctors predict stroke risks using some methods and their experience. But that may not give accurate results as required. Also they may lack in capturing complex relationship between various factors.

But now with the help of machine learning we can easily analyze large amount of medical data. We can also find complex relationship, patterns which can improve predictions. This can be done with the help of different machine learning models.

This study focuses on comparison between individual and ensemble machine learning models for predicting brain stroke. For that I have trained and tested four individual machine learning models Logistic Regression, Decision Tree, Random Forest and Gradient Boosting. After that all these four models are combine using soft voting to create an ensemble model. After checking performance of every model using different evaluation metrics the best performing model is selected to predict the stroke risk on unseen patient data.

II. LITERATURE REVIEW

Nowdays, machine learning is used to find out people who are at the risk of having a brain stroke. Many researchers and studies used different machine learning models and techniques to analyze data and predict stroke risks. In this section we will discuss studies on stroke prediction using machine learning and identify their findings and limitations.

Thakur & Gupta in their 2025 study used machine learning techniques to diagnose stroke early using clinical and demographic data. They used models like Logistic Regression, Decision Tree and Random Forest. They found that machine learning can effectively identify individuals at risk of stroke. The authors emphasized the importance of prediction to improve outcomes. They concluded that machine learning techniques Random Forest and Decision Tree are very good at predicting stroke risk and improving care. This study provides an understanding of how machine learning can be applied to predict stroke risk using machine learning. [1]

Mattas and colleagues in 2023 applied machine learning techniques a Decision Tree model, to support stroke prevention using clinical data. The authors highlighted the importance of analytics in identifying high-risk individuals. Emphasized the need for model interpretability in healthcare applications. The results showed that machine learning can help with decision-making.

However the study only evaluated one model. They did not compare individual and ensemble machine learning approaches using evaluation metrics. Additionally the model was not validated on data. [2]

Sharma and colleagues in 2024 machine learning classifiers for stroke risk assessment. Found that Logistic Regression was the performing model due to its accuracy and interpretability. This study highlighted the importance of choosing right models for healthcare domain. However this study did not use different learning techniques, models or even probability base risk prediction. [3]

Sailasya and Kumari (2021) studied few machine learning models. They got to know that Naïve Bayes is performing better compared to other models with good accuracy. This shows that simple models can also work well. However evaluating model only on accuracy is not enough in medical applications. [4]

Jain and Mangal (2024) found that Logistic Regression gave best result while studying on how data cleaning and tuning can improve the machine learning models. This shows that we can improve the performance of the models with good and accurate data processing. However their study only based on individual models and not on ensemble model. [5]

Kavitha and colleagues (2024) developed a stroke prediction system by comparing models against traditional classifiers based on different risk factors. Authors also addressed the issue of class imbalance with the help of SMOTE technique and also used Chi-Square, Information Gain to select important features. Their results says ensemble approach outperforms the baseline models with Random Forest achieving accuracy of 98.06%. [6]

Charmilisri and colleagues (2023) did research by using machine learning models to find out people who are at the risk of having stroke. They used people's health and lifestyle related data. They tested four models and those all performed well. Neural Network gave good predictions while Random Forest performed best with high sensitivity and specificity in predicting both high risk and low risk people. [7]

Ponselvakumar and colleagues (2022) aim to help in early prediction of stroke and developed a useful healthcare system using a web application. For this they used machine learning to predict the stroke risk using health related data. They used two models Decision Tree and Random Forest after cleaning and preparing the health related data. Their result shows that Random Forest performed better compared to Decision Tree with high accuracy of 94%. [8]

Wang and colleagues (2023) did research on the problem of handling unbalanced dataset while predicting stroke risk using methods like SMOTE, over-sampling and under-sampling. They found out that these methods improved the results, specially recall and ROC-AUC for models like Random Forest. Also they showed how accuracy alone is not ideal for medical data. To improve result in future, they suggest using more data and better methods. [9]

Saputra and colleagues (2023) did research on early prediction of who are at the risk of having a stroke with the help of machine learning models. For that they used patient's health data and applied different machine learning models to it after preprocessing. They compares the performance of these models using different evaluation metrics. Result shows that Random Forest model with good accuracy can support in early stroke prediction. Also this paper highlights the importance preprocessing with SMOTE and choosing suitable model for prediction. [10]

Tashkova and Colleagues in 2025 systematically evaluated machine learning models to identify the most effective algorithms for predicting stroke risk. Researchers evaluated five models. They found that SVM and Random Forest achieved the test accuracies. XGBoost demonstrated sensitivity when paired with SMOTE using machine learning. [11]

Islam and colleagues in 2021 proposed a machine learning-based framework for predicting stroke risk using data. Multiple classifiers were used. The Voting Classifier achieved the performance with approximately 98% accuracy. Future work can involve comparison using evaluation metrics and validation on unseen patient data using machine learning. [12]

Narmatha and colleagues in 2023 proposed an approach for stroke prediction using lifestyle-related data. The ensemble model achieved 95% accuracy outperforming classifiers. It also demonstrated recall and F1-score indicating identification of high-risk patients using machine learning. [13]

James and colleagues in 2024 applied learning techniques to enhance prediction accuracy and robustness. Stacking-based ensemble models achieved the performance with accuracy reaching up to 99% significantly outperforming individual classifiers. The results confirm that ensemble learning improves reliability and classification performance using machine learning. [14]

Arshad and Sharmila in 2024 reviewed machine learning techniques used for stroke prediction, including models and ensemble methods. The review concludes that ensemble and hybrid models generally outperform classifiers by improving accuracy, stability and generalization. However it notes a lack of comparison on data in many existing reviews using machine learning. [15]

From the reviewed literature it is clear that machine learning techniques are widely used for stroke risk prediction with individual models demonstrating baseline performance and ensemble methods showing robustness and accuracy. However there are some gaps that remain:

Many studies rely primarily on accuracy and lack evaluation using metrics such as ROC-AUC, precision-recall curves and cross-validation.

Limited attention is given to validating models on data, which's critical for real-world clinical deployment.

While complex ensemble methods are explored, simpler and more interpretable ensemble approaches are not sufficiently evaluated in a framework against high-performing individual models.

To address these gaps the present study conducts a comparison of ensemble machine learning models using robust evaluation metrics and applies the best-performing ensemble model to predict stroke risk on unseen data thereby enhancing reliability and practical applicability, for clinical decision support systems using machine learning.

III. METHODOLOGY

A. Research Objective

The main goal of this study is to make and test computer programs that can predict the risk of stroke. We want to see what information is related to stroke like age and health issues and use this to make predictions. We will try out computer programs to see which one works best. Then we will use a method called Voting Classifier to combine these programs and make our predictions more accurate. We will check how well each program works and use measurements to see how good they are. Finally we will use one of these computer programs to predict the risk of stroke in patients we have not seen before using the information we have, about them like their clinical and symptoms-based data to check if our stroke risk prediction is correct.

B. Research Design

This paper is based on an experimental machine learning research design to compare and evaluate the prediction ability of individual and ensemble models in stroke risk prediction. All the models are trained and tested on the same data using similar data preprocessing techniques.

C. Data Collection

The dataset used for this study was collected from Kaggle, which consists of 35,000 patient health records with 19 features associated with patient health conditions concerning stroke risk. The dataset consists of patient attributes like demographic features, health conditions, and symptoms concerning stroke risk. The dataset consists of two variables concerning outcome:

- **At Risk** – This is a target variable indicating whether a patient is at risk of a stroke.
- **Stroke Risk Percentage** – This is a continuous variable indicating a calculated risk value concerning patient symptoms and health indicators.

Since the stroke_risk_percentage variable is directly related to the target variable, it was removed during model training to avoid data leakage.

D. Feature Description

The dataset has features. These features can be put into three groups: demographic attributes, medical history attributes and symptom-based attributes.

Demographic Attributes:

Age - This is how old the patient is in years. Age is very important in predicting stroke because the older you get the more likely you are to have a stroke.

Gender - This is whether the patient is a man or a woman. We change this into a code before we use it.

Medical History Attributes and Clinical Conditions:

These are health issues related to the heart that can lead to a stroke. It include things like blood pressure, irregular heartbeats and heart pain.

Symptom-Based Attributes:

The dataset also contains information about symptoms that could indicate someone is at risk of a stroke. These symptoms are either present or not.

Target Variable:

At Risk - This indicates the patient is either at the risk of having a stroke (1) or not (0).

E. Selected ML Models

Individual Models:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting

Ensemble Model: Voting Classifier combining the four models above using soft voting.

F. Evaluation Metrics

These things are used to see how well a model works to see its performance:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

G. Implementation Tools

Programming Language: Python

Platform: Google Colab / Jupyter Notebook

Python Libraries: pandas, numpy, scikit-learn, xgboost, matplotlib, seaborn, joblib, etc.

H. Research Workflow

The flowchart below shows the essential steps of the research, from data loading to evaluating different models. Also comparison between both individual and ensemble models.

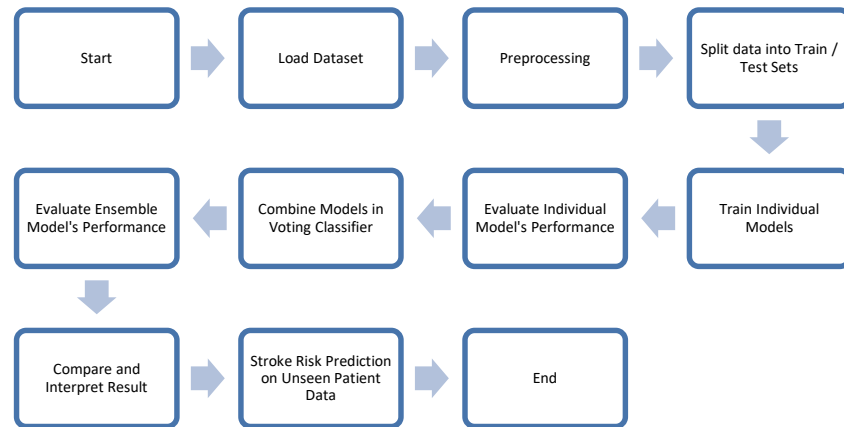


Figure: 1 Research Flowchart

IV. IMPLEMENTATION

A. Experimental Setup

The experimental implementation was carried out using Python and Google Colab. Scikit-learn was used for developing and evaluating the models. Data preprocessing and analysis were carried out using libraries such as pandas, numpy, matplotlib, and seaborn.

B. Exploratory Data Analysis (EDA)

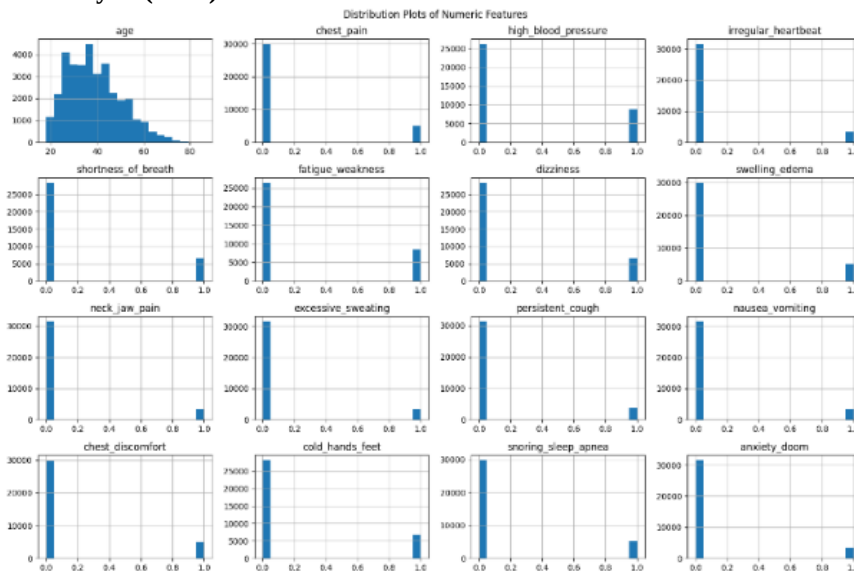


Figure: 2 Feature Distribution Plots

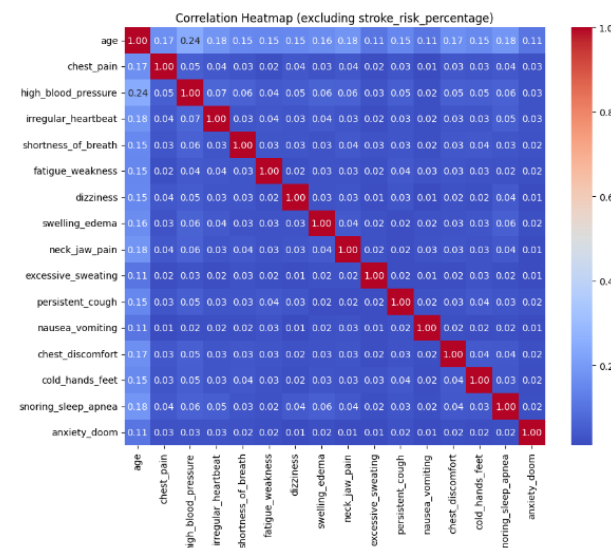


Figure: 3 Correlation Heatmap

The dataset was looked at first to see how the features are spread out and how they are connected to each other. To get a sense of the numbers distribution plots were used to see how the numerical features are spread out. It was found that most of the feature-related data was of binary nature, but the feature related to age showed a slightly skewed distribution.

A correlation heatmap was used to understand the relationship between the features, excluding the derived feature `stroke_risk_percentage`. Most of the features showed low to moderate correlation, indicating independence between the features.

C. Data Preprocessing

The overall workflow of data preprocessing used in this study is shown in Figure 4.

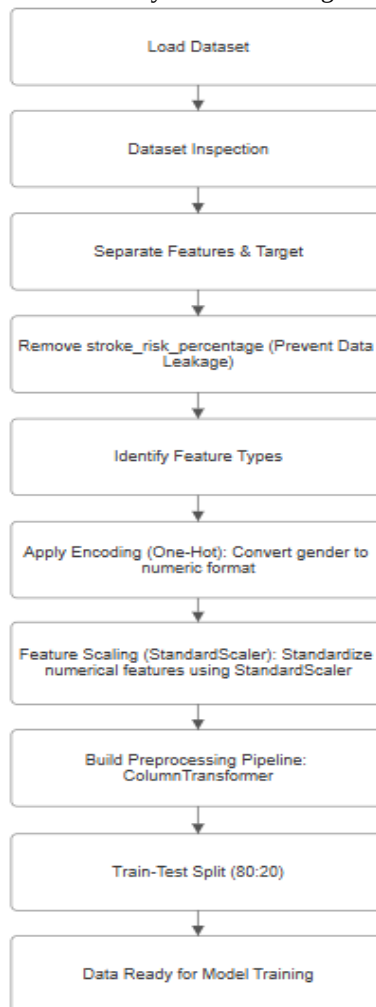


Figure: 4 Data Processing Workflow

After performing the exploratory analysis, the dataset was prepared for modeling. For this purpose, the method of one-hot encoding was used for the categorical variable "gender." For numerical variables, standard scaling was used with the help of "StandardScaler." Also, the derived variable "stroke_risk_percentage" was removed since it is related to the target variable. Finally, the dataset was split into training and testing datasets in the ratio of 80:20 using stratified sampling.

D. Model Development

Four individual machine learning models were implemented:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting

All four machine learning models were implemented using a pipeline approach consisting of both training and testing classification algorithms.

Five-fold cross-validation is used on the training set for evaluating the stability of the machine learning model.

E. Ensemble Model Construction

In order to improve the robustness of the predictions, an ensemble model was developed using a soft Voting Classifier.

The Voting Classifier combines the probabilities of the predictions of the individual models to arrive at a decision based on the average probability of the predictions.

F. Model Evaluation Strategy

Model performance was evaluated based on the following metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

ROC-AUC was given more importance as it is a threshold-independent performance metric for classification problems, which is critical in the case of medical risk prediction problems.

V. RESULT AND FINDINGS

A. Performance Comparison of Individual Models

Summary of Model Performance:								
	Model	Test Accuracy	Test Precision	Test Recall	Test F1 Score	Test ROC AUC	CV ROC AUC Mean	CV ROC AUC Std
0	Logistic Regression	0.957429	0.944596	0.939464	0.942023	0.992253	0.991599	0.000637
4	Voting Classifier	0.946429	0.949755	0.902212	0.925373	0.988591	0.987557	0.001018
2	Random Forest	0.925714	0.960591	0.832363	0.891892	0.984592	0.982719	0.000767
3	Gradient Boosting	0.927714	0.955185	0.843229	0.895713	0.981563	0.980194	0.001465
1	Decision Tree	0.894714	0.921245	0.780753	0.845201	0.952603	0.950511	0.003672

Figure: 5 Models Performance Table

The performance of individual machine learning models is provided in Table Logistic Regression had the highest accuracy among the individual models, which shows the classification ability of the model. Random Forest and Gradient Boosting have similar performance with good balance in precision and recall. Decision Tree has lower recall, which shows that the model has more misclassifications.

B. Confusion Matrix Analysis

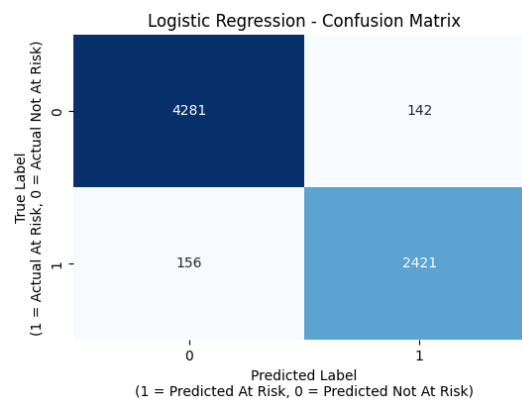


Figure: 6 Confusion Matrix (Logistic Regression)

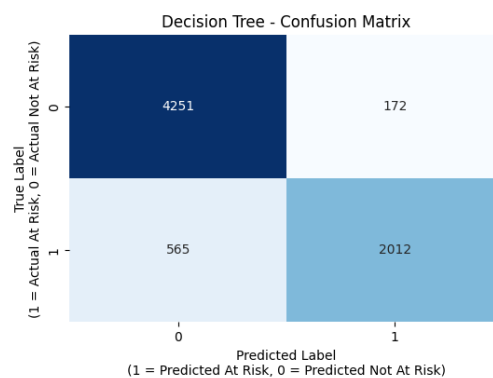


Figure: 7 Confusion Matrix (Decision Tree)

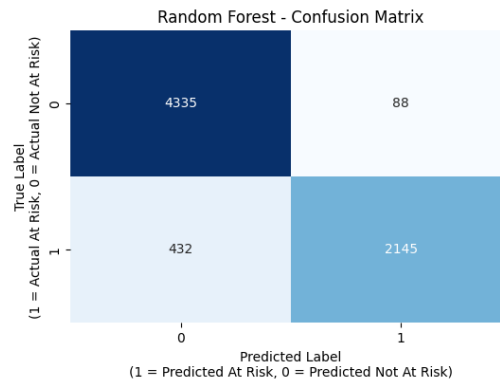


Figure: 8 Confusion Matrix (Random Forest)

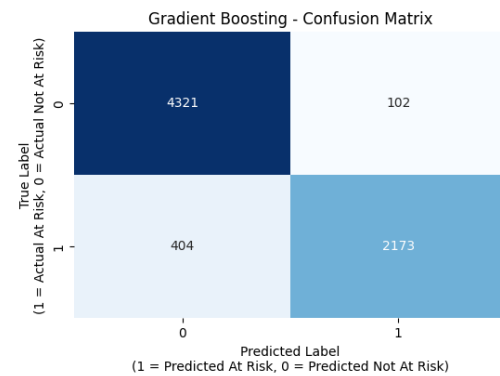


Figure: 9 Confusion Matrix (Gradient Boosting)

The confusion matrices were used to analyze the prediction behavior. Logistic Regression showed a strong balance of true positive and true negative predictions. Decision Trees resulted in higher false negative predictions than the other ensemble models. Random Forest and Gradient Boosting resulted in the reduction of misclassification errors.

C. Ensemble Model Performance

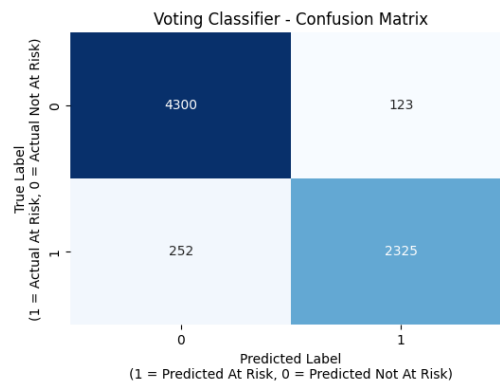


Figure: 10 Confusion Matrix (Voting Classifier)

The Voting Classifier combined the results of all the individual models using soft voting. The performance of the ensemble model was well-balanced in terms of accuracy, precision, recall, and F1-score. The ensemble method was better than the individual model methods in the sense that the results were more stable and less biased.

D. ROC-AUC Comparative Analysis

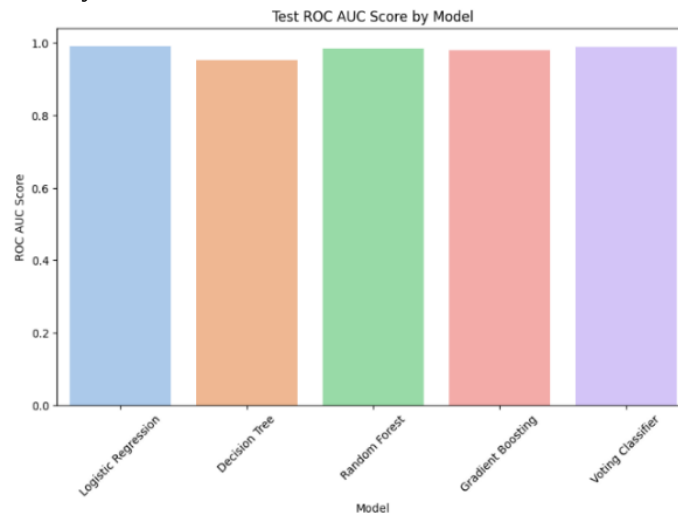


Figure: 11 ROC-AUC Curve Across all Models

Figure shows the ROC-AUC comparison of the models. Logistic Regression had the highest ROC-AUC score among all the models, which shows that the class separation was good. The Voting Classifier also showed good results in the ROC-AUC score, which confirms the effectiveness of ensemble learning methods on the robustness of the classifiers.

E. Validation on Unseen Data

In order to evaluate the applicability of the model, it was tested on unseen patient cases for a final evaluation. The ensemble model is able to predict both high-risk and low-risk scenarios. It also produces a stroke risk estimate based on the probabilities.

Case 1: High Risk of Stroke

```
Predicted At Risk: 1
Stroke Risk Probability: 0.9658
Stroke Risk Percentage (Model-estimated): 96.58 %
```

Figure: 12 Prediction (At Stroke Risk)

Case 2: Low Risk of Stroke

```
Predicted At Risk: 0
Stroke Risk Probability: 0.0751
Stroke Risk Percentage (Model-estimated): 7.51 %
```

Figure: 13 Prediction (Not at Stroke Risk)

VI. CONCLUSION

This study presented a comparative analysis of individual and ensemble machine learning models in predicting stroke risk using clinical and symptom-based data.

For individual machine learning model performance, the Logistic Regression model showed the best performance in terms of overall accuracy and class-wise performance. The soft voting classifier showed the best performance in the context of ensemble machine learning model performance.

The results showed the efficiency and effectiveness of machine learning model performance in analyzing clinical risk factors and predicting stroke risk at an early stage.

VII. FUTURE SCOPE

Although the proposed models demonstrated strong predictive performance using clinical and symptom-based data, several improvements can be explored in future research.

First, the study can be extended using real-world hospital datasets to improve clinical validation and enhance model generalizability across diverse patient populations.

Secondly, medical imaging data, for example, real-world CT scan images for stroke detection and risk assessment, can be incorporated in the future. By using a combination of clinical data and medical imaging data and deep learning techniques, for example, CNN models, the predictive results can be significantly improved.

Thirdly, advanced ensemble techniques, for example, a stacking framework or a boosting framework, for example, XGBoost, can be incorporated to improve the predictive results.

Lastly, cost-sensitive learning techniques can be used to improve the predictive results by reducing false negatives, which is a critical problem in medical risk prediction. Furthermore, the model can be extended to a real-time decision support tool for medical professionals to evaluate stroke risks.

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